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Teacher-guided AI-assisted development of a virtual single-slit diffraction laboratory for Grade 10 physics: a classroom case study

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Abstract

This classroom case study describes the teacher-guided AI-assisted development of a virtual single-slit diffraction laboratory for Grade 10 physics. A generative AI system was used as a rapid prototyping partner to create and iteratively refine an interactive simulation, on-screen measurement ruler, and supporting digital laboratory worksheet. The development process was shaped by repeated teacher feedback on classroom usability, conceptual accessibility, visual clarity, measurement precision, and error analysis. The final activity allowed students to vary wavelength, slit width, and screen distance, observe the resulting diffraction pattern, measure the central maximum, and compare measured values with the theoretical relation. Classroom implementation and exploratory student feedback are discussed in terms of engagement, visualization, usability, and perceived support for experimental understanding. The case illustrates how generative AI can support physics teachers in developing curriculum-linked virtual laboratory activities when the teacher remains central in directing the pedagogy, checking the physics, and adapting the tool for learners.

1 Introduction

Interactive simulations and virtual laboratories have become increasingly important tools in physics education, particularly for topics involving abstract processes, dynamic systems, or experiments that are difficult to reproduce consistently in school laboratories. Simulations can support conceptual visualization, inquiry-based learning, and rapid exploration of relationships between physical variables. However, the development of custom educational simulations often requires programming experience, substantial preparation time, and repeated testing to ensure that the final tool remains pedagogically suitable for classroom use.

Recent advances in generative artificial intelligence (AI) systems have introduced new possibilities for rapid prototyping of educational materials, including code generation, worksheet design, and interactive visualization tools. Despite growing interest in AI-assisted educational development, much of the discussion remains focused either on the technical capabilities of AI systems or on concerns surrounding their use in education. Comparatively fewer studies have examined how physics teachers may iteratively collaborate with generative AI systems to create classroom-oriented learning tools while retaining pedagogical control over the design process.

This paper presents two classroom case studies involving teacher-guided AI-assisted development of virtual physics laboratory activities for secondary education. In the first case, a Grade 10 single-slit diffraction simulation was iteratively developed to support visualization, measurement, and error analysis during a diffraction laboratory activity. In the second case, a Grade 12 transformer simulation was developed using a substantially shorter prompting process after design patterns and pedagogical expectations had already been established through the earlier interaction. Together, these cases provide insight into how generative AI may function as a rapid prototyping partner during simulation development while highlighting the continuing importance of teacher guidance, physics verification, and classroom adaptation.

Table 1. Caption text describing the table. Adapt the template table below or replace with a new table. To add more tables, copy and paste the whole `\begin{table}...\end{table}` block.

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Rather than treating AI as an autonomous educational designer, this study examines a collaborative workflow in which the teacher repeatedly evaluated outputs, corrected unsuitable features, refined the classroom usability of the simulations, and integrated the resulting tools into curriculum-linked learning activities. The paper focuses on the development workflow, classroom implementation, and exploratory student feedback associated with these AI-assisted virtual laboratories.

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Data availability

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Supplementary data

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References